Exploratory Data Analysis on Advertising Sales Dataset

Abstract

Exploratory Data Analysis (EDA) is an essential step in the data analysis process that helps in understanding the structure and characteristics of data. This project analyzes the Advertising Sales dataset to identify relationships between advertising expenditures across TV, Radio, and Newspaper channels and their impact on sales.

Various statistical techniques and visualization methods were used to discover trends, patterns, correlations, and influential factors. The findings indicate that TV and Radio advertising have a significant positive impact on sales, whereas Newspaper advertising shows relatively little influence.

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1. Introduction

Exploratory Data Analysis is the process of examining datasets to summarize their main characteristics using statistical methods and visualizations. EDA helps identify patterns, detect anomalies, test assumptions, and discover relationships between variables.

The Advertising Sales dataset contains information regarding advertising expenditure through different media channels and corresponding sales figures. This project aims to determine which advertising channels contribute most significantly to sales performance.

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2. Objectives

- To understand the structure and characteristics of the dataset.
- To perform data cleaning and preprocessing.
- To analyze statistical summaries.
- To identify relationships among variables.
- To visualize data distributions and correlations.
- To derive actionable business insights.

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3. Dataset Description

The dataset consists of 200 observations and four numerical features.

Feature	Description
TV	Advertising budget allocated to TV
Radio	Advertising budget allocated to Radio
Newspaper	Advertising budget allocated to Newspaper
Sales	Product sales generated

Dataset Source: Kaggle Advertising Dataset

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4. Tools and Technologies

- Python
- Jupyter Notebook
- Pandas
- NumPy
- Matplotlib
- Seaborn

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5. Methodology

The project was completed using the following steps:

1. Data collection
2. Data loading
3. Data inspection
4. Data cleaning
5. Statistical analysis
6. Data visualization
7. Correlation analysis
8. Insight generation
9. Conclusion and recommendations

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6. Data Preprocessing

The following preprocessing steps were performed:

- Checked dataset dimensions
- Examined data types
- Identified missing values
- Removed duplicate records
- Detected outliers using box plots
- Generated descriptive statistics

No missing values or duplicate records were identified in the dataset.

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7. Exploratory Data Analysis

Several statistical techniques were applied to understand the dataset.

Univariate Analysis

- Distribution of advertising budgets
- Distribution of sales values
- Detection of outliers

Bivariate Analysis

- TV vs Sales
- Radio vs Sales
- Newspaper vs Sales

Correlation Analysis

A correlation matrix was created to examine relationships between variables.

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8. Visualizations

The following visualizations were created:

- Histogram
- Box Plot
- Pair Plot
- Correlation Heatmap
- Scatter Plot: TV vs Sales
- Scatter Plot: Radio vs Sales
- Scatter Plot: Newspaper vs Sales

Include screenshots of all visualizations in this section.

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9. Key Insights

- TV advertising has the strongest positive correlation with sales.
- Radio advertising moderately influences sales.
- Newspaper advertising has minimal impact on sales.
- Higher spending on TV campaigns generally results in increased sales.
- The dataset contains no missing values or duplicate records.
- A few outliers were observed in Newspaper advertising expenditure.

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10. Conclusion

The analysis demonstrates that advertising expenditure significantly affects product sales. TV advertising emerged as the most influential factor, followed by Radio advertising. Newspaper advertising showed relatively weak correlation with sales.

Organizations should prioritize investment in TV and Radio channels to improve marketing effectiveness and maximize returns.

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11. Future Scope

Future enhancements to this project may include:

- Building predictive machine learning models
- Applying feature engineering techniques
- Comparing multiple regression algorithms
- Performing advanced statistical analysis
- Evaluating model performance metrics

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12. References

1. Pandas Documentation: https://pandas.pydata.org/
2. NumPy Documentation: https://numpy.org/
3. Matplotlib Documentation: https://matplotlib.org/
4. Seaborn Documentation: https://seaborn.pydata.org/
5. Kaggle Advertising Dataset
